Grade 7 Practice

Standard 7.RP.1

Name: _____ Date: _____

Computation

1. A car goes $\frac{1}{2}$ mile in $\frac{1}{4}$ hour. What is its speed in miles per hour?

_____

2

---

2. $\frac{3}{4}$ pound of cheese costs $3. What is the price per pound? _____

$4

---

3. $\frac{2}{3}$ cup of juice is poured in $\frac{1}{2}$ hour. How many cups per hour? _____

$\frac{4}{3}$

---

4. 1 $\frac{1}{2}$ miles is walked in $\frac{1}{2}$ hour. What is the speed in miles per hour?

_____

3

Word problems

5. A snail moves $\frac{2}{5}$ inch in $\frac{1}{10}$ minute. How many inches per minute is that?

Show your work and answer.

Answer: 4 inches per minute

---

6. Juice uses 3 $\frac{1}{2}$ cups for 7 people. How many cups per person?

Show your work and answer.

Answer: $\frac{1}{2}$ cup per person

Grade 7 Practice

Standard 7.RP.2

Name: _____ Date: _____

Computation

1. Is y = 3x a proportional relationship? Write yes or no.

yes

2. In a table, x = 2 gives y = 6 and x = 5 gives y = 15. What is the constant of proportionality? _____

3

3. Is y = x + 2 a proportional relationship? Write yes or no.

no

4. Items cost $4 each. What is the constant of proportionality? _____

4

Word problems

5. Pay is $9 for 3 hours. What is the constant of proportionality in dollars per hour?

Show your work and answer.

Answer: 3

6. A line passes through (0, 0) and (4, 10). What is the constant of proportionality?

Show your work and answer.

Answer: 2.5

Grade 7 Practice

Standard 7.RP.3

Name: _____ Date: _____

Computation

1. What is 25% of 80? _____ **20**

2. What is a 15% tip on $40? _____ **6**

3. A $50 item is 20% off. What is the sale price? _____ **40**

4. A $200 amount increases by 10%. What is the new amount? _____ **220**

Word problems

5. A shirt costs $32. Sales tax is 8%. What is the total cost?

Show your work and answer.

Answer: $34.56

6. A class has 30 students, and 40% are boys. How many boys are in the class?

Show your work and answer.

Answer: 12 boys

Grade 7 Practice

Standard 7.NS.1

Name: _____ Date: _____

Computation

1. Solve: -3 + 8 = _____

5

2. Solve: 5 - (-2) = _____

7

3. Solve: -4 + (-6) = _____

-10

4. Solve: -7 - 2 = _____

-9

Word problems

5. The temperature is -5° and rises 9°. What is the new temperature?

Show your work and answer.

Answer: 4°

6. An account balance is -$12. Then $20 is deposited. What is the new balance?

Show your work and answer.

Answer: $8

Grade 7 Practice

Standard 7.NS.2

Name: _____ Date: _____

Computation

1. Solve: $(-3)(4) =$ _____

-12

2. Solve: $(-6)(-5) =$ _____

30

3. Solve: $-24 \div 6 =$ _____

-4

4. Solve: $(-\frac{1}{2})(\frac{2}{3}) =$ _____

$-\frac{1}{3}$

Word problems

5. A diver goes down 5 meters every minute for 4 minutes. What is the total change in depth?

Show your work and answer.

Answer: -20 meters

6. Temperature drops 3° each hour for 6 hours. What is the total change?

Show your work and answer.

Answer: -18°

Grade 7 Practice

Standard 7.NS.3

Name: _____ Date: _____

Computation

1. Solve: -2.5 + 1.75 = _____

-0.75

2. Solve: $\frac{3}{4} \times (-\frac{4}{9}) =$ _____

$-\frac{1}{3}$

3. Solve: -1.2 × (-5) = _____

6

4. Solve: $-\frac{1}{2} + (-\frac{1}{4}) =$ _____

$-\frac{3}{4}$

Word problems

5. Ana's score is -2.5 points. She then gains 4.75 points. What is her new score?

Show your work and answer.

Answer: 2.25 points

6. A diver is at -12.5 meters and rises 3.75 meters. What is the new depth?

Show your work and answer.

Answer: -8.75 meters

Grade 7 Practice

Standard 7.EE.1

Name: _____ Date: _____

Computation

1. Expand: 3(x + 4) = _____

3x + 12

2. Simplify: 2x + 3x - x = _____

4x

3. Expand: -2(y - 5) = _____

-2y + 10

4. Factor: 6x + 9 = _____

3(2x + 3)

Word problems

5. Simplify: 5n + 2 + 3n - 1

Show your work and answer.

Answer: 8n + 1

6. Simplify: 7 - 2(a + 3)

Show your work and answer.

Answer: 1 - 2a

Grade 7 Practice

Standard 7.EE.2

Name: _____ Date: _____

Computation

1. Is 4(x + 3) equivalent to 4x + 12? Write yes or no. **yes**

2. Is 2x + 5 + x equivalent to 3x + 5? Write yes or no. **yes**

3. Is 3(x − 2) equivalent to 3x − 2? Write yes or no. **no**

4. Write 5x − 10 in factored form. _____ **5(x − 2)**

Word problems

5. Simplify 0.5x + 0.25x.

Show your work and answer.

Answer: 0.75x

6. An item priced at p is 20% off. Write an expression for the sale price.

Show your work and answer.

Answer: 0.8p

Grade 7 Practice

Standard 7.EE.3

Name: _____ Date: _____

Computation

1. Solve: 2x + 5 = 17. x = _____ **6**

2. Solve: 3(x - 4) = 9. x = _____ **7**

3. Solve: x/4 + 2 = 6. x = _____ **16**

4. Solve: -2x + 3 = 11. x = _____ **-4**

Word problems

5. A phone plan costs $15 plus $0.10 per text. Your total bill is $25. How many texts did you send?

Show your work and answer.

Answer: 100 texts

6. Sam earns $9 per hour and pays an $8 fee. He nets $46. How many hours did he work?

Show your work and answer.

Answer: 6 hours

Grade 7 Practice

Standard 7.EE.4

Name: _____ Date: _____

Computation

1. Solve: $x + 4 > 9$. $x >$ _____ **5**

2. Solve: $2x < 10$. $x <$ _____ **5**

3. Solve: $-3x \geq 12$. $x \leq$ _____ **-4**

4. Solve: $4(x + 1) = 20$. $x =$ _____ **4**

Word problems

5. A bike costs at most $150. Sam pays $40 down and then $25 each month. What is the greatest number of whole months he can pay?

Show your work and answer.

Answer: 4 months

6. Two numbers add up to 18. The second is 2 more than the first. What is the first number?

Show your work and answer.

Answer: 8

Grade 7 Practice

Standard 7.G.1

Name: _____ Date: _____

Computation

1. On a map, 1 cm = 5 m. What is 4 cm in meters? _____

20

2. A drawing has a scale of 1:100. A 6 cm line is how many cm in real life?

_____

600

3. A scale is 2 inches = 10 feet. What is 3 inches in feet? _____

15

4. A scale is 1 cm : 2 km. What is 5 cm in km? _____

10

Word problems

5. A floor plan uses 1 inch for 3 feet. A room is 4 inches on the plan. How long is the real room?

Show your work and answer.

Answer: 12 feet

6. A model car is built at a 1:24 scale. The real car is 96 inches long. How long is the model?

Show your work and answer.

Answer: 4 inches

Grade 7 Practice

Standard 7.G.2

Name: _____ Date: _____

Computation

1. Can sides 3, 4, and 5 form a triangle? Write yes or no. **yes**

2. Can sides 2, 3, and 6 form a triangle? Write yes or no. **no**

3. A triangle has three 60° angles. What type of triangle is it? **equilateral**

4. A triangle has angles 90° and 40°. What is the third angle? _____ **50°**

Word problems

5. A triangle has angles of 30° and 70°. What is the third angle?

Show your work and answer.

Answer: 80°

6. Two sides of a triangle are 5 and 7. Can the third side be 1? Write yes or no.

Show your work and answer.

Answer: no

Grade 7 Practice

Standard 7.G.3

Name: _____ Date: _____

Computation

1. A cut parallel to the base of a cube makes what shape? _____

square

2. A cylinder is sliced straight down the middle. What shape is the cross section? _____

rectangle

3. A sphere is sliced in any direction. What shape is the cross section? _____

circle

4. A triangular prism is cut parallel to its base. What shape is the cross section? _____

triangle

Word problems

5. A cone is sliced horizontally. What shape is the cross section?

Show your work and answer.

Answer: circle

6. A cylinder is sliced straight down the middle. What shape is the cross section?

Show your work and answer.

Answer: rectangle

Grade 7 Practice

Standard 7.G.4

Name: _____ Date: _____

Computation

1. A circle has radius 3. Using π ≈ 3.14, find its area. _____ **28.26**

2. A circle has diameter 10. Using π ≈ 3.14, find its circumference. _____ **31.4**

3. A circle has radius 5. Using π ≈ 3.14, find its circumference. _____ **31.4**

4. A circle has diameter 4. Using π ≈ 3.14, find its area. _____ **12.56**

Word problems

5. A pizza has a radius of 7 inches. Using π ≈ 3.14, what is its area?

Show your work and answer.

Answer: **153.86 square inches**

6. A wheel has a diameter of 2 feet. Using π ≈ 3.14, what is its circumference?

Show your work and answer.

Answer: **6.28 feet**

Grade 7 Practice

Standard 7.G.5

Name: _____ Date: _____

Computation

1. What is the supplement of a 110° angle? _____ **70°**

2. What is the complement of a 35° angle? _____ **55°**

3. Vertical angles are equal. If one is 50°, what is the other? _____ **50°**

4. Two angles form a straight line. One is 65°. What is the other? _____ **115°**

Word problems

5. Two angles of a triangle are 50° and 60°. What is the third angle?

Show your work and answer.

Answer: 70°

6. Angles 120° and x form a straight line. What is x?

Show your work and answer.

Answer: 60°

Grade 7 Practice

Standard 7.G.6

Name: _____ Date: _____

Computation

1. A triangle has base 12 and height 5. What is its area? _____ **30**

2. A cube has side length 4. What is its volume? _____ **64**

3. A rectangular prism is 5 by 2 by 3. What is its volume? _____ **30**

4. A cube has side length 2. What is its surface area? _____ **24**

Word problems

5. A box is 3 feet by 4 feet by 5 feet. What is its volume in cubic feet?

Show your work and answer.

Answer: 60 cubic feet

6. A wall is 10 feet by 8 feet. How many square feet of paint cover it?

Show your work and answer.

Answer: 80 square feet

Grade 7 Practice

Standard 7.SP.1

Name: _____ Date: _____

Computation

1. Which sample is random? Write A or B. A) Ask 5 friends. B) Choose 20 names from a full class list at random.

B

2. Is a sample chosen at random usually representative of the population? Write yes or no.

yes

3. A survey uses only volunteers. Is this a random sample? Write yes or no.

no

4. A sample is chosen by drawing names from a hat. Is it random? Write yes or no.

yes

Word problems

5. Mia asks only the first 10 people who arrive at school. Is her sample random? Write yes or no.

Show your work and answer.

Answer: **no**

6. Lee randomly picks 30 students from a group of 300. How many students are in the population?

Show your work and answer.

Answer: **300 students**

Grade 7 Practice

Standard 7.SP.2

Name: _____ Date: _____

Computation

1. A sample of 40 has 10 with a trait. Estimate the number in a population of 400. _____

100

2. In a sample of 10, 3 like red. Estimate for 500 people. _____

150

3. 25% of a sample of 200 students like math. About how many students like math? _____

50

4. In a sample of 60, 15 say yes. What is the fraction of yes answers? _____

$\frac{1}{4}$

Word problems

5. A survey of 100 people shows 60% like pizza. About how many of 2,000 people like pizza?

Show your work and answer.

Answer: 1,200 people

6. 8 of 20 students in a sample like the new lunch menu. About how many of 500 students like it?

Show your work and answer.

Answer: 200 students

Grade 7 Practice

Standard 7.SP.3

Name: _____ Date: _____

Computation

1. Group A has a mean of 10 and group B has a mean of 12. Which mean is greater? Write A or B.

B

2. Group A has a range of 5 and group B has a range of 9. Which group has more spread? Write A or B.

B

3. Group A has a mean of 20 and group B has a mean of 15. What is the difference of the means? _____

5

4. Which has a greater spread: a range of 4 or a range of 10? Write the range.

10

Word problems

5. Class A has a mean score of 80 and class B has a mean of 72. How many points higher is class A's mean?

Show your work and answer.

Answer: 8 points

6. Class A has a range of 10 and class B has a range of 4. Which class has greater spread?

Show your work and answer.

Answer: Class A

Grade 7 Practice

Standard 7.SP.4

Name: _____ Date: _____

Computation

1. Sample A has a mean of 70 and sample B has a mean of 65. What is the difference? _____

5

2. Sample A has a range of 12 and sample B has a range of 6. What is the difference in range? _____

6

3. Population A has a median of 40 and population B has a median of 35. Which has the higher median? Write A or B.

A

4. Means are 70 and 71, and each group has a range of 20. Is the difference meaningful? Write yes or no.

no

Word problems

5. Two shops have mean prices of $12 and $15. What is the difference in the means?

Show your work and answer.

Answer: $3

6. Group A has a median height of 150 cm and group B has a median of 160 cm. Which group is taller? Write A or B.

Show your work and answer.

Answer: B

Grade 7 Practice

Standard 7.SP.5

Name: _____ Date: _____

Computation

1. What is the probability of rolling a 4 on a fair die? _____

$\frac{1}{6}$

2. What is the probability of flipping heads on a fair coin? _____

$\frac{1}{2}$

3. What is the probability of an impossible event? _____

0

4. What is the probability of a certain event? _____

1

Word problems

5. A bag has 10 marbles and 3 are red. What is the probability of picking a red marble?

Show your work and answer.

Answer: $\frac{3}{10}$

6. In the same bag, what is the probability of not picking a red marble?

Show your work and answer.

Answer: $\frac{7}{10}$

Grade 7 Practice

Standard 7.SP.6

Name: _____ Date: _____

Computation

1. A coin is flipped 100 times and lands heads 60 times. What is the relative frequency of heads? _____

0.6

2. A spinner is spun 50 times and lands on blue 10 times. What is the relative frequency? _____

0.2

3. A die is rolled 60 times and a six comes up 10 times. What is the relative frequency? _____

$\frac{1}{6}$

4. An event happens 5 times in 20 trials. What is the estimated probability? _____

0.25

Word problems

5. A student tosses a cup 40 times and it lands upright 15 times. What is the relative frequency of landing upright?

Show your work and answer.

Answer: **0.375**

6. A relative frequency is 0.4 after 100 trials. How many successes were there?

Show your work and answer.

Answer: **40**

Grade 7 Practice

Standard 7.SP.7

Name: _____ Date: _____

Computation

1. A fair die is rolled. What is P(even)? _____

$\frac{1}{2}$

2. A model gives each of outcomes 1 to 6 equal probability. What is P(5)?

_____

$\frac{1}{6}$

3. Two fair coins are flipped. What is P(two heads)? _____

$\frac{1}{4}$

4. A spinner has 8 equal sections. What is the probability of one section?

_____

$\frac{1}{8}$

Word problems

5. A bag has 4 red and 6 blue marbles. What is the probability of picking blue?

Show your work and answer.

Answer: $\frac{3}{5}$

6. Three outcomes have probabilities 0.3, 0.5, and 0.1. If these are the only outcomes, what must the probability of a missing fourth outcome be? (Hint: the total must be 1.)

Show your work and answer.

Answer: **0.1**

Grade 7 Practice

Standard 7.SP.8

Name: _____ Date: _____

Computation

1. Two dice are rolled. What is P(sum is 7)? _____

$\frac{1}{6}$

2. Two coins are flipped. What is P(heads then tails)? _____

$\frac{1}{4}$

3. A card is drawn from a standard deck. What is P(red or ace)? _____

$\frac{7}{13}$

4. Two cards are drawn with replacement. What is P(both red)? _____

$\frac{1}{4}$

Word problems

5. A bag has 3 red and 2 blue marbles. A marble is picked, not replaced, then another. What is P(red then blue)?

Show your work and answer.

Answer: $\frac{3}{10}$

6. Two cards are drawn with replacement from a bag of 3 red and 2 blue cards. What is P(red then red)?

Show your work and answer.

Answer: $\frac{9}{25}$